# Computational Modeling of Interfacial Stability

## Chemical Stability

The chemical reaction energies between each cathode material and the five halide solid electrolytes (SEs) were computed using the `InterfacialReactivity` module in `pymatgen`, with thermodynamic data retrieved from the Materials Project database. Figure 1 and Figure 2 show the chemical reaction energy as a function of the cathode molar fraction for $FeF_3$ and $FeCl_3$, respectively. The molar fraction $x$ is defined as $x = n_{\text{cathode}}/(n_{\text{cathode}} + n_{\text{SE}})$, and the reaction energy is normalized per atom of the reactant mixture. A negative value indicates thermodynamic instability at that composition.

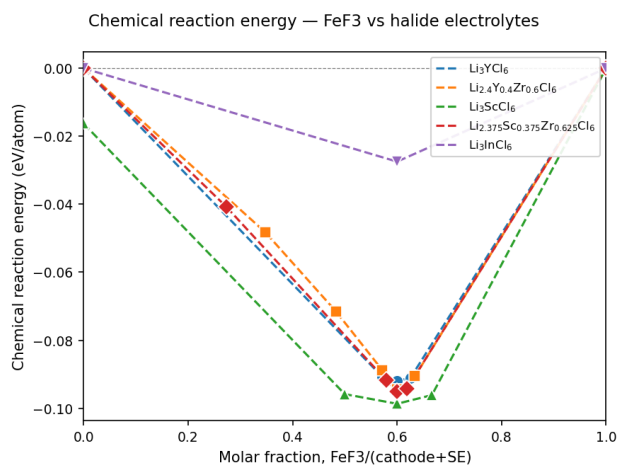


Figure 1: Chemical reaction energy between $FeF_3$ and five halide solid electrolytes as a function of cathode molar fraction. Markers indicate kink compositions where the decomposition products change.

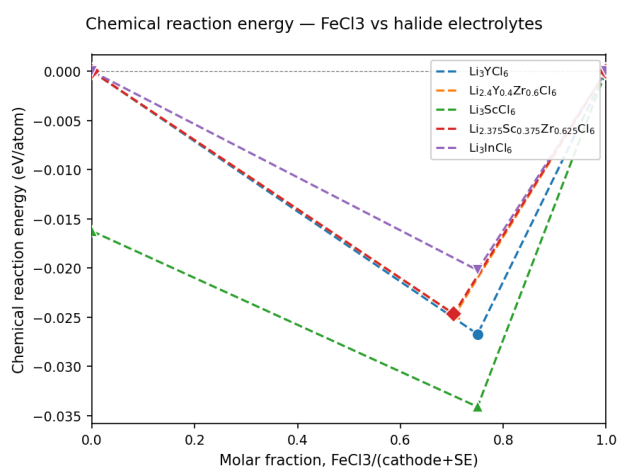


Figure 2: Chemical reaction energy between $FeCl_3$ and five halide solid electrolytes as a function of cathode molar fraction.

## Electrochemical Stability

The electrochemical stability was evaluated using the grand-potential phase diagram (GPCD) formalism, in which Li is treated as an open element exchanged with a reservoir at chemical potential $\mu_{\mathrm{Li}} = \mu_{\mathrm{Li}}^{\mathrm{ref}} - V$, where $V$ is the electrode voltage versus $\mathrm{Li/Li^+}$. The reaction energy was computed across the voltage window of 1.0–4.2 V in steps of 0.01 V, with curves colored from blue (1.0 V) to red (4.2 V) following a coolwarm colormap. Figures 3 and 4 show the resulting voltage-dependent reaction energy profiles for $FeF_3$ and $FeCl_3$, respectively.

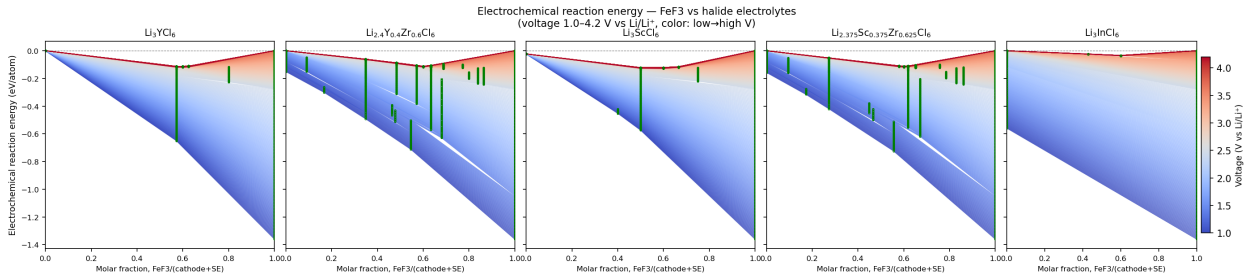


Figure 3: Electrochemical reaction energy between $FeF_3$ and each of the five halide solid electrolytes at voltages from 1.0 V (blue) to 4.2 V (red) vs. $\mathrm{Li/Li^+}$. Green hollow circles mark kink positions where the stable decomposition products change.

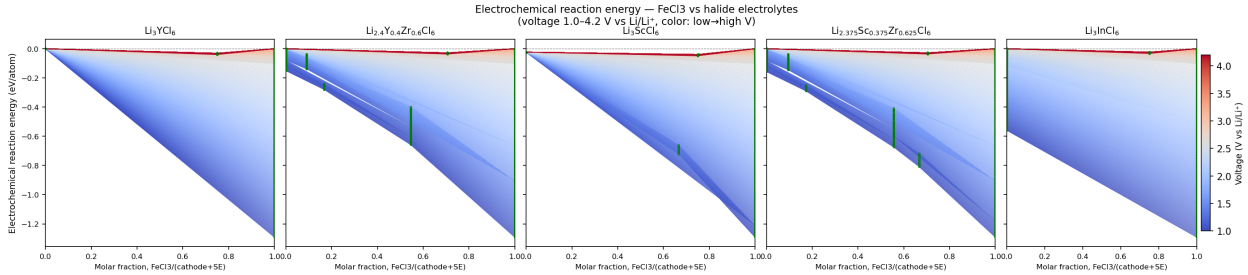


Figure 4: Electrochemical reaction energy between $FeCl_3$ and each of the five halide solid electrolytes at voltages from 1.0 V (blue) to 4.2 V (red) vs. $\mathrm{Li/Li^+}$.

## Mixed Cathode: $FeF_3$+$FeCl_3$ (1:1 Molar)

For the mixed cathode composed of $FeF_3$ and $FeCl_3$ in a 1:1 molar ratio, two computational approaches were employed. In the first approach (Option A), the mixture is represented as a single pseudo-compound $Fe_2F_3Cl_3$, and the binary `GrandPotentialInterfacialReactivity` pipeline is applied identically to the single-cathode cases above (Figures 5 and 6).

In the second approach (Option C), $FeF_3$ and $FeCl_3$ are treated as two independent reactant phases. The reaction energy is computed by scanning the ternary composition line $c(x) = \frac{x}{2}\,\mathrm{FeF_3} + \frac{x}{2}\,\mathrm{FeCl_3} + (1-x)\,\mathrm{SE}$ using the GPCD hull energy directly, with the reference energy defined as the weighted sum of the individual hull energies of $FeF_3$, $FeCl_3$, and SE (Figures 7 and 8). This approach captures reactions between $FeF_3$ and $FeCl_3$ themselves, and reveals additional phase boundaries not visible in Option A.

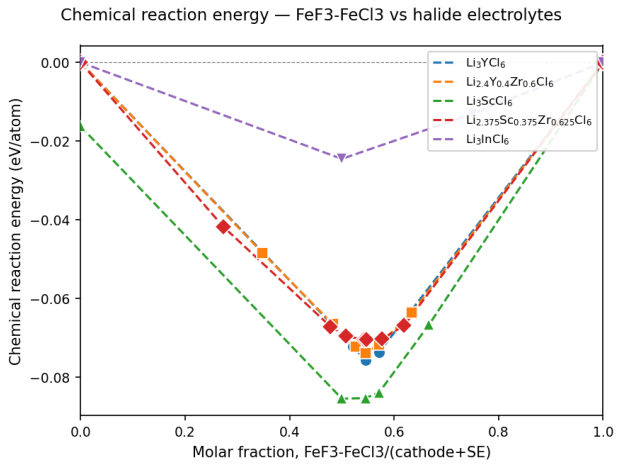


Figure 5: Chemical reaction energy for the pseudo-compound $Fe_2F_3Cl_3$ (representing a 1:1 molar $FeF_3$+$FeCl_3$ mixture) vs. five halide solid electrolytes (Option A).

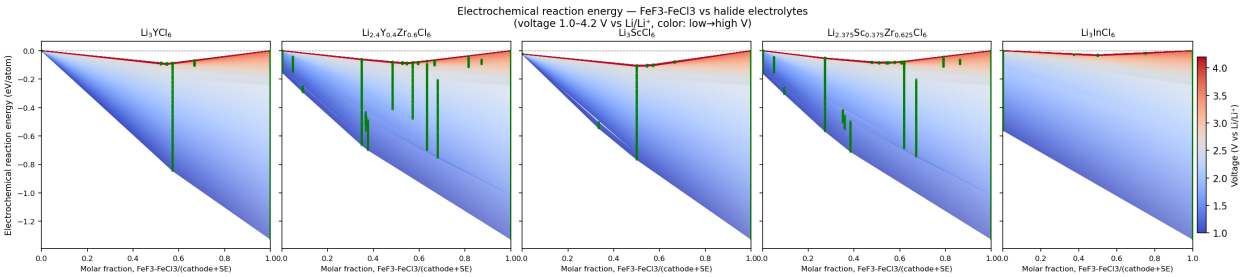


Figure 6: Electrochemical reaction energy for $Fe_2F_3Cl_3$ vs. five halide solid electrolytes at 1.0–4.2 V vs. $Li/Li^+$ (Option A).

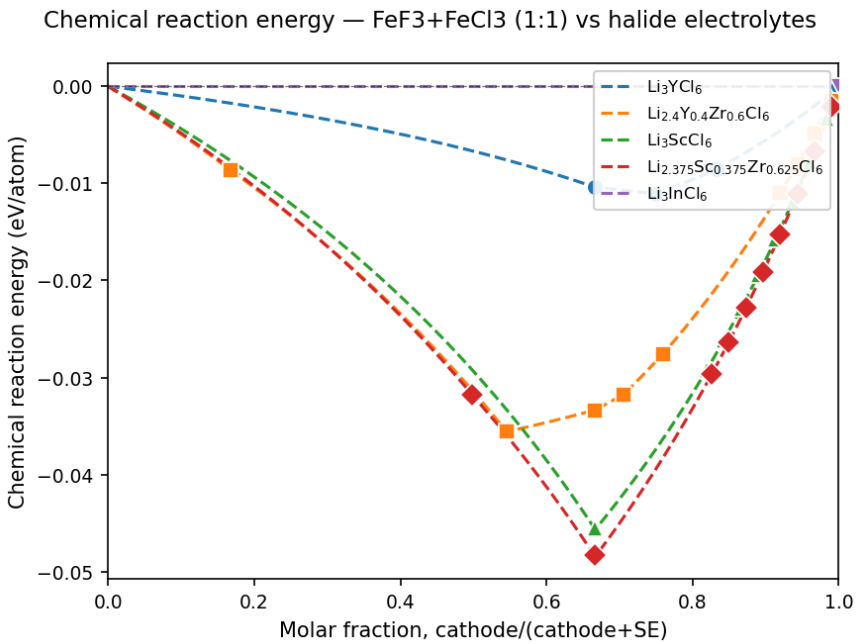


Figure 7: Chemical reaction energy for the ternary $FeF_3$+$FeCl_3$+SE system vs. five halide solid electrolytes (Option C). $FeF_3$ and $FeCl_3$ are treated as separate reactant phases.

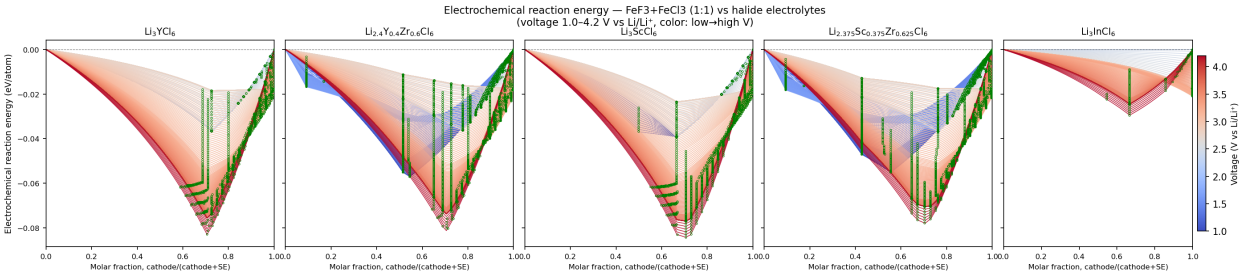


Figure 8: Electrochemical reaction energy for the ternary system at 1.0–4.2 V vs. $Li/Li^+$ (Option C).